



Project Title:

Diabetes Prediction Using a Feature-Level Multimodal Fusion Network.

Course: CSE400A

Submitted by:

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1. Research Area Identification

The research area of this capstone project is artificial intelligence in healthcare, with a strong focus on applying deep learning techniques for disease detection and prediction. In recent years, artificial intelligence has played an important role in improving medical diagnosis by analyzing complex and large-scale data. This project specifically aims to predict diabetes at an early stage, which is critical for preventing serious health complications. To achieve this, the system analyzes multiple types of medical data rather than relying on a single data source. The project combines eye fundus images, which can reveal visual signs related to diabetes and diabetic complications, with patient clinical information such as age, blood glucose level, body mass index, blood pressure, and other relevant health indicators. By integrating both visual and numerical data, the proposed system is designed to capture complementary information from different sources. This multimodal approach helps improve prediction reliability and accuracy compared to traditional systems that use only images or only clinical data.

Descriptive Title

The selected project title is:

“Diabetes Prediction Using a Feature-Level Multimodal Fusion Network.”

This title has been carefully chosen to clearly represent the core idea of the project. It highlights that the system is designed for diabetes prediction and that it uses a multi-modal approach, meaning more than one type of data is involved. The phrase *feature-level fusion network* indicates that features extracted from different data sources are combined within the model to make the final prediction. As a result, the title provides a clear understanding of both the methodology and the objective of the project.

Avoidance of Abbreviations

No abbreviations have been used in the project title to ensure clarity and simplicity. Using complete and meaningful terms makes the title easy to read and understand for a wide audience. This includes instructors, evaluators, and readers who may not have a strong technical background in artificial intelligence or medical data analysis. Avoiding abbreviations also helps maintain academic professionalism and improves the overall quality of the capstone report.

Reflection of Scope and Purpose

The chosen title accurately reflects the scope and purpose of the project. It clearly indicates that the project focuses on predicting diabetes by integrating multiple data sources through a feature level fusion network. The title communicates that the system is not limited to a single type of data or a simple model, but instead uses an intelligent and structured approach to improve prediction performance.

Overall, the title effectively represents the project’s goal of developing a robust and intelligent system that supports early and accurate diabetes prediction, which can assist healthcare professionals in making better clinical decisions.

2. Research Problem

2.1 Problem Definition

Diabetes is one of the fastest-growing chronic diseases worldwide, and diabetic nephropathy (DR) is among its most severe complications. DR damages retinal blood vessels and can lead to irreversible vision loss or blindness if not detected and treated early. According to global health statistics, a significant proportion of diabetic patients develop some form of DR during their lifetime, making early screening and accurate diagnosis critically important.

Despite advances in medical imaging and artificial intelligence, current diabetic nephropathy screening systems face several limitations. Traditional DR diagnosis relies heavily on manual inspection of fundus images by trained ophthalmologists. This process is time-consuming, costly, and prone to inter-observer variability. In many developing countries, including Bangladesh, the shortage of trained retinal specialists and limited access to screening facilities further aggravate the problem, especially in rural areas. Recent studies have introduced machine learning (ML) and deep learning (DL) techniques for automated DR detection and diabetes risk prediction. While these approaches demonstrate high accuracy in controlled experimental settings, many of them suffer from limited generalization, lack of explainability, small or biased datasets, and insufficient real-world validation.

2.2 Why the Problem Needs a Solution

The importance of addressing this problem is multifaceted:

1. Late Diagnosis Leads to Blindness

DR often remains asymptomatic in its early stages. Without routine screening, patients are frequently diagnosed only after irreversible retinal damage has occurred.

2. Healthcare Resource Constraints

Manual screening places a heavy workload on ophthalmologists. In low resource settings, the ratio of specialists to patients is extremely low, making mass screening impractical.

3. Inconsistent Diagnostic Decisions

Human grading of fundus images can vary significantly between experts, leading to inconsistent diagnoses and treatment delays.

4. Limited Clinical Trust in AI Systems

Many high-performing AI models operate as black boxes, offering little insight into how predictions are made. This lack of transparency reduces clinical adoption.

5. Fragmented Research Focus

Some studies focus only on binary DR detection, others on multiclass grading, and some on diabetes risk prediction but few integrate accuracy, explainability, robustness, and real-world applicability together.

2.3 Who Is Affected

- **Diabetic Patients:** Risk of vision loss or blindness; kidney disease; nerve damage (nephropathy); heart disease; stroke; reduced quality of life; mental stress.
- **Ophthalmologists & Healthcare Professionals:** Higher workload; risk of errors and fatigue.
- **Rural & Low-Income Communities:** Limited access to care; delayed diagnosis; financial burden.

2.4 Who Is Affected

- **Diabetic Patients:** Risk of vision loss or blindness; kidney disease; nerve damage (nephropathy); heart disease; stroke; reduced quality of life; mental stress.
- **Ophthalmologists & Healthcare Professionals:** Higher workload; risk of errors and fatigue.
- **Rural & Low-Income Communities:** Limited access to care; delayed diagnosis; financial burden.
- **Healthcare Systems:** Increased treatment costs; strain on resources.
- **Families & Caregivers:** Emotional and financial stress; increased caregiving responsibilities.
- **Policy Makers & Public Health Authorities:** Challenges in implementing screening programs; need for funding and AI-based tools.

2.5 Research Gap Identification

Based on the analysis of the eight reviewed papers, the following research gaps are identified:

2.5.1 Lack of Unified Systems

Most studies address either DR detection from images or diabetes risk prediction from tabular data, but very few integrate multimedia data with explain ability.

2.5.2 Limited Explain ability (XAI)

Although some papers use SHAP or CAM-based methods, many high-accuracy models lack meaningful clinical explanations.

2.5.3 Over-Reliance on Augmentation and Small Datasets

Several studies report extremely high accuracy but rely heavily on data augmentation or small datasets, raising concerns about overfitting.

2.5.3 Insufficient Real-World Validation

Many models are evaluated on single datasets without cross-bencher or prospective clinical validation.

2.5.5 Binary Classification Dominance

Binary classification oversimplifies disease progression and does not support clinical decision-making effectively.

3. Background

3.1 Introduction to the Topic and Context

Diabetes is one of the most common and rapidly increasing chronic diseases worldwide. Changes in lifestyle, unhealthy eating habits, and lack of physical activity have contributed to a significant rise in diabetes cases. If diabetes is not detected at an early stage, it can lead to serious complications such as heart disease, kidney failure, nerve damage, and vision problems. Therefore, early prediction and detection of diabetes are extremely important for effective treatment and prevention of long-term health issues.

In recent years, artificial intelligence and deep learning have been widely used in the healthcare sector to improve disease detection and diagnosis. These technologies can analyze large and complex medical data, including images and clinical records, to support accurate and timely medical decision-making.

3.2 Detailed Explanation of the Problem

Most existing diabetes prediction systems rely on a single type of data. Some studies focus only on clinical data such as age, blood glucose level, body mass index, and blood pressure, while others rely only on medical images, such as eye fundus images, to identify diabetes-related complications. However, using only one data source often provides an incomplete understanding of a patient's health condition, which can reduce prediction accuracy.

Medical images and clinical data each contain different but complementary information. When these data types are analyzed separately, valuable insights may be missed. This limitation highlights the need for a system that can combine multiple data sources to provide a more comprehensive and reliable diabetes prediction.

3.3 Supporting Evidence from Literature

Previous studies have shown that deep learning models, particularly convolutional neural networks, perform well in extracting meaningful features from fundus images related to diabetes and diabetic retinopathy. Similarly, machine learning and neural network models have demonstrated strong performance in analyzing clinical and numerical patient data. However, most of the reviewed research uses image data and clinical data independently. Only a limited number of studies explore feature-level fusion techniques that integrate both data types within a single predictive framework. As a result, there remains a clear research gap in developing a unified multimodal system that effectively combines image and clinical data for diabetes prediction.

3.4 Importance of the Study

This capstone project addresses the identified research gap by proposing a feature-level multimodal fusion network that integrates eye fundus images with clinical data for diabetes prediction. By combining multiple data sources, the proposed system aims to improve prediction accuracy, robustness, and reliability. The outcomes of this study are expected to support healthcare professionals by providing an intelligent decision-support tool for early diabetes prediction. Early identification of diabetes can help initiate timely treatment, reduce complications, and improve overall patient health outcomes.

4. Literature Review

This section reviews existing research on diabetic retinopathy detection and diabetes risk prediction using machine learning and deep learning techniques.

4.1 Deep Learning-Based DR Detection Systems

Zhang et al. (2022) proposed a multidimensional deep learning platform aligned with the UK NHS diabetic retinopathy screening guidelines. Their system uses five classifiers and ensemble CNN architectures, achieving high AUROC values. While the study demonstrates strong robustness and clinical alignment, it remains limited to binary classification and lacks full DR staging. Shamrat et al. (2024) introduced DRNet13, a lightweight CNN designed for multi-class DR classification. The model emphasizes computational efficiency and outperforms several pretrained CNNs. However, the study depends on a single dataset and lacks advanced explainability techniques. Tymchenko et al. (2020) proposed a multi-head ensemble CNN architecture combining classification, regression, and ordinal regression. The model achieved exceptionally high kappa scores, indicating strong agreement with expert graders. Despite its strong performance, the approach is computationally complex and lacks real-world deployment analysis.

4.2 Region-Specific and Real-World Validation Studies

Haque et al. (2021) conducted one of the first Bangladesh-specific DR validation studies, demonstrating high accuracy on real clinical images collected from Bangladeshi hospitals. The study highlights the importance of geographic generalization but suffers from a very small validation dataset and absence of explainable.

4.3 Multimodal and Ensemble-Based Approaches

Rahman et al. (2025) introduced a multimodal DR severity assessment system combining ultra-widefield fundus images with clinical tabular data. The use of Integrated Gradients and SHAP significantly improves interpretability. However, overall classification accuracy remains moderate, suggesting room for improvement. Rajeshwar et al. (2025) proposed a hybrid attention-based stacking ensemble, combining deep feature extraction with classical ML classifiers. Although the model achieves extremely high accuracy, the small dataset and lack of XAI reduce its reliability for clinical deployment.

4.4 Diabetes Risk Prediction Using Machine Learning

Sun et al. (2025) conducted a large-scale diabetes risk prediction study using over 556,000 patient records. XGBoost with SHAP explainability outperformed all other models, demonstrating the importance of large datasets and interpretability. Verma et al. (2025) proposed a CatBoost–SMOTEENN–Optuna pipeline, achieving nearperfect ROC-AUC scores. Despite excellent performance, the study lacks explainability and external validation.

4.5 Summary of Literature Gaps

From the reviewed studies, the following limitations are consistently observed:

- 4.5.1 Limited integration of image and tabular data
- 4.5.2 Inadequate explainability for clinical trust
- 4.5.3 Small or single-source datasets
- 4.5.4 Over-simplified classification strategies
- 4.5.5 Lack of deployment-oriented evaluation

These gaps justify the need for a capstone project that focuses on accuracy, robustness, explainability, and real-world applicability simultaneously.

4.6 Methodology:

In paper Automated multimodal severity assessment of diabetic retinopathy using ultra-widefield color fundus photography and clinical tabular data. Exploratory Data Analysis (EDA) was performed to ensure data reliability and to gain meaningful insights prior to model development. The image dataset was examined for resolution, aspect ratio, and overall quality to maintain consistency across ultra-widefield fundus images. Random visualization aided in understanding retinal patterns and disease-related variations. In parallel, statistical distribution analysis of clinical tabular features was conducted to identify data imbalance, outliers, and underlying trends. Additionally, SHAP-based feature importance analysis was employed to interpret the influence of clinical variables on diabetic retinopathy severity, supporting informed feature integration and enhancing model transparency before training.

In paper Automated multidimensional deep learning platform for referable diabetic retinopathy detection. Exploratory Data Analysis (EDA) was carried out to thoroughly understand the characteristics of the fundus image dataset before model training. Class distribution analysis was performed to identify potential class imbalance among referable and non-referable diabetic retinopathy cases. Image quality grading analysis helped assess variations in illumination, focus, and noise that could affect feature extraction. Furthermore, t-SNE visualization was employed to explore deep feature embeddings and verify class separability in the learned feature space. To enhance interpretability, CAM and SHAP-CAM-based visual inspections were used to highlight lesion-focused regions, thereby improving model reliability and clinical trustworthiness.

In paper An advanced deep neural network for fundus image analysis and enhancing diabetic retinopathy detection. Exploratory Data Analysis (EDA) was conducted to evaluate the dataset characteristics and support effective model learning. Dataset size and class distribution were analyzed to identify imbalance issues and ensure balanced training. Sample fundus images were visualized before and after preprocessing to verify enhancement of relevant retinal structures and noise reduction. The impact of data augmentation techniques was examined to confirm improved data diversity and robustness. Additionally, feature map visualization from intermediate CNN layers was performed to gain insight into how the network progressively extracts discriminative retinal features, thereby validating the model's learning behavior and interpretability.

In paper Deep Learning Approach to Diabetic Retinopathy Detection. Exploratory Data Analysis (EDA) was performed to assess data consistency and model behavior prior to final evaluation. Label distribution analysis was conducted to identify potential class imbalance and labeling inconsistencies within the dataset. Sample-level prediction comparisons were used to examine differences in model outputs across individual samples. Additionally, ensemble output consistency was inspected to evaluate robustness and agreement among

multiple prediction heads. Test-time augmentation (TTA) behavior analysis further helped understand prediction stability under augmented inputs, ensuring reliable and consistent diabetic retinopathy classification performance.

In paper Diabetic Retinopathy Detection Using DL-Based Feature Extraction and a Hybrid Attention Based Stacking Ensemble. Exploratory Data Analysis (EDA) was conducted to understand dataset characteristics and support reliable ensemble learning. Dataset balance analysis between diabetic retinopathy and normal images was performed to identify potential imbalance and overfitting risks due to limited data size. The effects of preprocessing were visualized to confirm enhancement of salient retinal structures. Feature extractor output analysis was carried out to assess the quality and discriminative power of deep learning-based representations. Finally, comparative performance analysis across different classifiers helped validate the suitability of extracted features for hybrid attention-based stacking ensemble models.

In paper Convolutional Nets for Diabetic Retinopathy Screening in Bangladeshi Patients. Exploratory Data Analysis (EDA) was performed to investigate dataset variability and real-world deployment challenges. Cross-dataset distribution comparison between Indian and Bangladeshi fundus images was conducted to analyze domain shift and population-specific differences. Image quality variation analysis helped assess inconsistencies arising from acquisition conditions. The impact of data augmentation was evaluated to ensure improved generalization across datasets. Additionally, error distribution analysis across diabetic retinopathy stages was carried out, revealing that most misclassifications occurred within ± 1 severity level, which is considered clinically acceptable and supports the model's practical screening reliability.

In paper Machine learning-based assessment of diabetes risk. Exploratory Data Analysis (EDA) was conducted to ensure data quality and reliability for diabetes risk prediction. Missing value analysis was performed to identify incomplete records, followed by outlier detection using the interquartile range (IQR) method to reduce noise and extreme bias. Feature distribution visualizations helped understand the statistical behavior and scale of clinical variables. Class imbalance analysis was carried out to mitigate biased learning toward majority classes. Additionally, SHAP-based feature importance analysis was employed to interpret model predictions and identify the most influential risk factors contributing to diabetes assessment.

In paper An Intelligent Machine Learning Pipeline for Early Diabetes Prediction: CatBoost Ensemble with SMOTEENN and Optuna Tuning. Exploratory Data Analysis (EDA) was conducted to examine data quality and class distribution prior to model development. Visualization of class imbalance highlighted the skewed nature of the dataset and the need for resampling. Outlier analysis helped identify noisy and extreme instances that could negatively affect learning. Feature encoding inspection was performed to verify proper representation of categorical and numerical variables. Additionally, before-and-after resampling distribution analysis was carried out to evaluate the effectiveness of SMOTEENN in improving minority class representation, thereby enhancing model generalization and predictive stability.

4.7 Comparative Performance Analysis of Models

Title	Author's	Model	Accuracy
Automated multimodal severity assessment of diabetic retinopathy using ultra-widefield colorfundus photography and clinical tabular data	Alireza Rezaei, Sarah Matta, Rachid Zeghlache, Pierre-Henri Conze, Capucine Lepicard, Pierre Deman, Laurent Borderie, Deborah Cosette, Sophie Bonnin, Aude Couturier, Béatrice Cochener, Mathieu Lamard, Mostafa El-Habib Daho, Gwenolé Quéllec	MLP + ResNet50	63.4%
A Hybrid Machine Learning Approach for Enhanced Diabetes Prediction: Integrating Image and Numerical Data	Ahmad Shaker Abdalrada, Ali Fahem Neamah, Huda Lafta Majeed, Ahmed Raad Al-Sudani	CNN + GBM	96%
Explainable Hybrid Deep Learning Framework with Multimodal Inputs for Diabetic Retinopathy Detection	Prasanna Sahu, Ashwani Kumar, Rituraj Jain, Kamal Upreti, Dileep Kumar Yadav, G. V. Radhakrishnan	MLP + Attention Fusion	99.34%
Diabetic Retinopathy Detection and Classification Using Mixed Models for a Disease Grading Database	Anas Bilal, Guangmin Sun, Yu Li, Sarah Mazhar, Abdul Qadir Khan	Support Vector Machine (RBF kernel)	98.06%
Diabetic Retinopathy Detection Using DL-Based Feature Extraction and a Hybrid Attention-Based Stacking Ensemble	Sanjana Rajeshwar, Shreya Thapliyal, Anbarasi M., Siva Shanmugam G.	Hybrid Stacking Ensemble	99.768%
A multimodal deep learning architecture for predicting interstitial glucose for effective type 2 diabetes management	Muhammad Salman Haleem, Daphne Katsarou, Eleni I. Georga, George E. Dafoulas, Alexandra Bargiota, Laura Lopez-Perez, Miguel Rujas, Giuseppe Fico, Leandro Pecchia, Dimitrios Fotiadis, Gatekeeper Consortium	Multimodal BiLSTM + CNN + Baseline Fusion	96.7%

Leveraging Deep Learning and Multi-Modal Data for Early Prediction and Personalized Management of Type 2 Diabetes	Shashi Bhushan Singh, Dr. Arjun Singh	CNN for time-series data (CGM, lifestyle) MLP for static data (EHR, genetic) Late fusion strategy with attention mechanisms Sigmoid output layer for T2DM risk prediction	89.7%
A deep learning based model for diabetic retinopathy grading	Samia Akhtar, Shabib Aftab, Oualid Ali, Munir Ahmad, Muhammad Adnan Khan, Sagheer Abbas, Taher M. Ghazal	RSG-Net (2- Stage)	99.37%
Automated multidimensional deep learning platform for referable diabetic retinopathy detection	Guihua Zhang, Jian-Wei Lin, Ji Wang, Jie Ji, Ling-Ping Cen, Weiqi Chen, Peiwen Xie, Yi Zheng, Yongqun Xiong, Hanfu Wu, Dongjie Li, Tsz Kin Ng, Chi Pui Pang, Mingzhi Zhang.	CNN ensemble (Inception-V3 + Xception + Inception-ResNet-V2)	97%
Early Detection and Severity Classification of Diabetic Retinopathy Using Convolutional Neural Networks	S. A. Karthik, M. N. Geetha, K. Prabhavathi, Dhananjaya Shashank, K. P. Suhaas, M. Narender	Lightweight CNN (Custom CNN architecture)	Binary Classification Accuracy: 96.2% Early Detection Accuracy: 85.2% Severity Classification Accuracy: 79.4%
An advanced deep neural network for fundus image analysis and enhancing diabetic retinopathy detection	F. M. Javed Mehedi Shamrat, Rashiduzzaman Shakil, Sharmin, Nazmul Hoque Ovy, Bonna	DRNet13	99%

	Akter, Md Zunayed Ahmed, Kawsar Ahmed, Francis M. Bui, Mohammad Ali Moni		
Distribution of diabetic retinopathy in diabetes mellitus patients and its association rules with other eye diseases	Xi Yao, Xiaoting Pei, Yingrui Yang, Hongmei Zhang, Mengting Xia, Ranran Huang, Yuming Wang, Zhijie Li	No ML model (Statistical study)	N/A
Predicting diabetes using supervised machine learning algorithms on E-health records	Sulaiman Afolabi, Nurudeen Ajadi, Afeez Jimoh, Ibrahim Adenekan	KNN	96.09%
Predictive Machine Learning Models for Early Diabetes Diagnosis: Enhancing Accuracy and Privacy with Federated Learning	Zill E Huma, Nimra Tariq, Shaharyar Zaidi	XGBoost	90%
Diabetes prediction using machine learning and explainable AI techniques	Isfazzaman Tasin, Tansin Ullah Nabil, Sanjida Islam, Riasat Khan	XGBoos+ADASYN	81%
Development of Various Diabetes Prediction Models Using Machine Learning Techniques	Juyoung Shin, Jaewon Kim, Chanjung Lee, Joon Young Yoon, Seyeon Kim, Seungjae Song, Hun-Sung Kim	Model-2 (62 variable)	86.7%
Prediction and diagnosis of future diabetes risk: a machine learning approach	Roshan Birjais, Ashish Kumar Mourya, Ritu Chauhan, Harleen Kaur	Gradient Boosting	86%
Clinically adaptable machine learning model to identify early appreciable features of diabetes	Nurjahan Nipa, Mahmudul Hasan Riyad, Shahriare Satu, Walliullah, Koushik Chandra Howlader, Mohammad Ali Moni	Extra Trees (ET) – SDHD	97.11%
Hyperparameter-Tuned Machine Learning Techniques for the Prediction of Diabetes Status Among Bangladeshi Adults	Lakshmi Rani Kundu, Ajit Kumar Majumder	GBM	74.4%
Machine learning-based	Qi Sun, Xin Cheng, Kuo	XGBoost	97.5%

assessment of diabetes risk	Han, Yichao Sun, He Ren, Ping Li		
An Intelligent Machine Learning Pipeline for Early Diabetes Prediction: CatBoost Ensemble with SMOTTEEN and Optuna Tuning	Vishal Verma, Satish Kumar, Vandna Rani Verma, Alka Agrawal	CatBoost-Optuna	98.93%
Deep Learning for the Detection and Classification of Diabetic Retinopathy with an Improved Activation Function	Usharani Bhimavarapu, Gopi Battineni	ResNet-152	99.41%
Convolutional Nets for Diabetic Retinopathy Screening in Bangladeshi Patients	Ayaan Haque, Ipsita Sutradhar, Mahziba Rahman, Mehedi Hasan, Malabika Sarker	DenseNet121	93.02%
Selective Diabetic Retinopathy Screening with Accuracy-Weighted Deep Ensembles and Entropy-Guided Abstention	Jophy Lin	Ensemble (Filtered)	99.44%
Classification of Diabetic Retinopathy using PreTrained Deep Learning Models	Inas Al-Kamachy, Reza Hassanpour, Roya Choupani	Inception ResNetV2	68%
Deep Learning Approach to Diabetic Retinopathy Detection	Borys Tymchenko, Philip Marchenko, Dmitry Spodarets	Ensemble Model	99.3%
Deep Learning-Based Diabetic Retinopathy Detection	Fahman Saeed, Muhammad Hussain, Hatim A Aboalsamh, Fadwa Al Adel, Adi Mohammed Al Owaifeer	CNN Model	90%
Diabetes Detection from Non-Diabetic Retinopathy Fundus Images Using Deep Learning Methodology	Yovel Rom, Rachelle Aviv, Gal Yaakov Cohen, Yehudit Eden Friedman, Tsontcho Ianchulev, Zack	CNN (Proposed Model) – Per Patient Visit	79%

5. Research Questions

5.1 Problem Statement

Early and accurate prediction of diabetes remains a major healthcare challenge, particularly when relying solely on clinical tabular data or single-modality diagnostic indicators. While clinical parameters such as glucose level and BMI provide strong metabolic signals, they may fail to capture long-term physiological effects of diabetes. Conversely, retinal fundus images can reveal diabetes-related ocular manifestations, but these signs are not always present in early-stage patients. Relying on a single data modality therefore limits predictive robustness and interpretability. This project addresses this limitation by proposing an explainable feature-level multimodal fusion approach that integrates clinical and retinal information for improved diabetes prediction.

5.2 Main Research Question

Research Question 1:

Does a feature-level multimodal fusion of clinical tabular data and retinal fundus images improve diabetes prediction performance compared to clinical-only and image-only models?

5.2 Sub-Research Questions

Research Question 2:

How effectively can retinal fundus images provide supportive evidence for diabetes prediction when combined with clinical data?

Research Question 3:

How does the proposed multimodal fusion model perform across different clinical scenarios, such as early-stage diabetes, chronic indicators, and non-diabetic cases?

Research Question 4:

Does the incorporation of explainability techniques (Grad-CAM and SHAP) enhance the interpretability and clinical trustworthiness of the model's predictions?

5.3 Alignment with Objectives

The formulated research questions are directly aligned with the project objectives by:

- Evaluating the effectiveness of multimodal fusion in diabetes prediction.
- Assessing the contribution of retinal imaging as a supportive modality.
- Investigating the interpretability and clinical relevance of the proposed AI framework

6. Research Objectives

6.1 Overall Aim

The primary aim of this capstone project is to develop and evaluate a hybrid clinical– ophthalmic AI system that integrates both fundus image analysis and patient clinical data to accurately predict diabetes risk and assess diabetic retinopathy severity. The system seeks to combine the strengths of image-based deep learning and tabular data machine learning, providing a unified, interpretable, and robust solution suitable for clinical decision support and early intervention.

6.2 Specific Objectives

1. **Develop separate pipelines** for fundus image analysis and clinical (tabular) data using state-of-the-art deep learning and machine learning models.
2. **Design a multimodal fusion framework** that effectively integrates image-based and tabular-based features to improve predictive accuracy for diabetes risk and diabetic retinopathy severity.
3. **Implement explainable AI (XAI) techniques**, such as SHAP for tabular data and GradCAM for image data, to make the hybrid system interpretable and clinically trustworthy.
4. **Evaluate model performance** across both single-modality and multimodal pipelines using clinically relevant metrics: accuracy, sensitivity, specificity, F1-score, ROC-AUC, and Cohen’s kappa.
5. **Compare multiple modeling approaches** (image-only, tabular-only, and fusion models) to determine the most effective strategy for unified diabetes and DR assessment.
6. **Optimize the hybrid model** for computational efficiency to ensure it is practical for real world clinical deployment, including low-resource or rural settings.
7. **Assess generalizability and robustness** across different datasets and populations to ensure the model performs well under varied conditions.
8. **Provide actionable clinical insights** by highlighting key predictive features from both images and patient data to assist healthcare professionals in early diagnosis and treatment planning.

6.3 Alignment with Research Questions

Each objective directly addresses the research gaps identified in Step 2 and the limitations highlighted in Step 4. The inclusion of explainability, multiclass prediction, and robustness ensures that the project contributes meaningfully to both academic research and practical healthcare applications.

7. Models Used in the Multi-modal Fusion Framework

Data Type	Model Used	Purpose / Role
CSV (Fusion)	MLP (Multi-Layer Perceptron)	Extraction of clinical feature embedding for multi-modal fusion
CSV (Baseline)	Logistic Regression, Random Forest, CatBoost	Performance comparison for clinical-only diabetes prediction
Image (Fusion)	EfficientNet-B0 (CNN)	Extraction of retinal image features related to diabetes
Image (Baseline)	ResNet, MobileNet	Optional comparison for imaginably prediction
Fusion Mechanism	Feature-level concatenation	Integration of clinical and retinal feature vectors
Final Classifier	Fully Connected (Dense) Layer	Final diabetes prediction (Diabetic / Non-Diabetic)
Explainability	Grad-CAM, SHAP	Interpretation of image and clinical model decisions

8. Formal Decision Scenarios in the Multi-modal Fusion Model

Clinical (CSV) Evidence	Retinal (Image) Evidence	Model Interpretation	Final Decision
Strong	Normal	High metabolic risk without visible retinal damage	Diabetic (Early-stage)
Strong	Strong	Consistent clinical risk with retinal damage	Diabetic (Established / Chronic)
Weak / Borderline	Strong	Retinal damage suggests long-standing diabetes despite borderline clinical values	Likely Chronic Diabetes / High-Risk
Weak / Borderline	Normal	Mild or emerging risk without retinal manifestation	At-Risk / Pre-diabetic
Normal	Normal	No significant metabolic or retinal evidence	Non-Diabetic

9. Plan Data Collection and Analysis

In paper “*Automated Multimodal Severity Assessment of Diabetic Retinopathy Using Ultrawidefield Color Fundus Photography and Clinical Tabular Data*”, the study utilizes two complementary data sources: ultra-widefield color fundus images and structured clinical tabular data. Retinal images were collected during routine ophthalmic examinations, while clinical and laboratory information was extracted from electronic health records and accurately linked to the corresponding images. Convolutional Neural Networks were used for image analysis, and traditional machine learning models were applied to tabular data. Multimodal feature fusion techniques integrated both data types, and explainability methods such as SHAP and Integrated Gradients were employed to interpret model predictions. The research was conducted as a quantitative clinical study in a hospital setting.

In paper “*Automated Multidimensional Deep Learning Platform for Referable Diabetic Retinopathy Detection*”, large-scale fundus image datasets were collected from multiple hospitals and national diabetic retinopathy screening programs to ensure data diversity. Images were captured using standard retinal cameras and annotated by experienced ophthalmologists following NHS screening guidelines. Deep learning-based CNN architectures and ensemble learning methods were applied to enhance detection performance. Model interpretability and feature exploration were supported using t-SNE, CAM, and SHAP-CAM visualizations. This quantitative, guideline-driven clinical screening study aims to support automated referral decisions in real-world screening programs.

In paper “*An Advanced Deep Neural Network for Fundus Image Analysis and Enhancing Diabetic Retinopathy Detection*”, publicly available diabetic retinopathy fundus image datasets were used. Images were collected from open-access repositories and processed using noise removal, contrast enhancement, and extensive data augmentation techniques. Multiple pre-trained CNN models were evaluated alongside a proposed lightweight CNN architecture (DRNet13). Feature map visualization techniques were employed to understand the learned retinal features. The study follows a quantitative experimental design focused on improving detection accuracy while reducing computational complexity.

In paper “*Deep Learning Approach to Diabetic Retinopathy Detection*”, publicly available benchmark retinal image datasets were utilized for diabetic retinopathy classification. Images underwent preprocessing steps including noise removal, resizing, and data augmentation. A multi-head CNN ensemble framework was applied, incorporating both classification and regression heads. Test-time augmentation was used during inference to improve robustness. This quantitative experimental study focuses on enhancing prediction stability and consistency through ensemble deep learning techniques.

In paper “*Diabetic Retinopathy Detection Using DL-Based Feature Extraction and a Hybrid Attention-Based Stacking Ensemble*”, the Ocular Disease Intelligent Recognition (ODIR) dataset was used, containing both diabetic retinopathy and normal fundus images. Images were extracted and augmented to address limited sample size, with preprocessing steps such as filtering and contrast enhancement applied. Deep learning models including ResNet, Inception, and VGG were used for feature extraction, and the extracted features were combined with traditional machine learning classifiers using a stacking ensemble strategy. This quantitative experimental study was conducted in a controlled environment to evaluate ensemble-based detection performance.

In paper “*Convolutional Nets for Diabetic Retinopathy Screening in Bangladeshi Patients*”, the APTOS 2019 dataset and real-world fundus images collected from Bangladeshi hospitals were used. Images were

captured using both standard and handheld fundus cameras and labeled by qualified ophthalmologists to ensure clinical accuracy. DenseNet-based CNN models were trained and evaluated across datasets, with data augmentation and cross-domain validation techniques applied to assess generalization. This quantitative clinical validation study aims to develop low-cost and deployable diabetic retinopathy screening solutions for rural Bangladesh.

In paper “*Machine Learning-Based Assessment of Diabetes Risk*”, large-scale physical examination and lifestyle-related data were collected from a district-level hospital. The data were extracted from electronic medical records and cleaned using missing value handling, outlier detection, and normalization techniques. Statistical feature selection methods, multiple machine learning classifiers, SMOTE for class imbalance handling, and SHAP for explainability were applied. This quantitative population-based study focuses on early prediction of diabetes risk using structured healthcare data.

In paper “*An Intelligent Machine Learning Pipeline for Early Diabetes Prediction: CatBoost Ensemble with SMOTEENN and Optuna Tuning*”, a publicly available diabetes prediction dataset obtained from Kaggle was used. Data preprocessing included duplicate removal, categorical feature encoding, and SMOTEENN resampling to address class imbalance. Ensemble machine learning approaches were applied, with CatBoost serving as the primary model. Optuna was used for automated hyperparameter tuning to improve model performance. This research follows a quantitative machine learning approach aimed at developing a robust and generalizable early diabetes prediction pipeline.

10. Define Expected Results

10.1 Anticipated Outcomes

The proposed feature-level multimodal fusion framework is expected to achieve more robust and reliable diabetes prediction compared to single-modality approaches. By integrating clinical tabular data and retinal fundus images, the model is anticipated to:

- Improve overall prediction performance (Accuracy, F1-score, and ROC-AUC) over CSV-only and image-only models.
- Enable **early-stage diabetes detection**, particularly in cases where clinical indicators are present without visible retinal damage.
- Provide **stronger confidence in chronic diabetes**
- **Identification** when retinal abnormalities are observed alongside clinical risk factors.
- Reduce misclassification caused by noisy or incomplete single-modality data through complementary multi-modal evidence.

10.2 Project Deliverables

The expected deliverables of this project include:

1. A trained multimodal fusion model combining an MLP-based clinical feature extractor and a CNN-based retinal feature extractor.
2. Baseline models for CSV-only and image-only diabetes prediction for comparative evaluation.
3. A comprehensive performance comparison demonstrating the benefit and limitations of multimodal fusion.
4. Explainability outputs, including:
 - Grad-CAM visualizations highlighting retinal regions contributing to predictions.
 - SHAP-based feature importance plots identifying key clinical risk factors.
5. A complete project report documenting methodology, results, analysis, and conclusions.
6. Visual artifacts such as workflow diagrams, tables, and result charts suitable for academic presentation.

10.3 Practical and Academic Impact

10.3.1 Practical Impact

- The system can support clinical decision-making by assisting healthcare professionals in identifying diabetic patients using both clinical and retinal evidence.
- It has potential application in early screening programs, especially in resourceconstrained settings where comprehensive testing may not always be available.
- The explainable nature of the model increases trust and transparency, making it suitable for real-world medical deployment.

10.3.2 Academic Impact

The project demonstrates a structured application of multimodal learning in medical AI, contributing to existing research on diabetes prediction. It provides a comparative analysis highlighting when multimodal fusion is beneficial and when unimodal models may suffice. The work offers a reproducible framework that can be extended to other diseases or additional modalities, such as laboratory time-series data or OCT images.

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12.Appendix:

CO	CO Description	PO	How PO was Addressed	Learning Domain
CO1	Integrate new and previously acquired knowledge for identifying a real-life complex computer science and engineering problem as the capstone project.	PO1: Engineering Knowledge	EP1 – Defining the Engineering Problem: Applied theoretical knowledge of AI, deep learning, and multimodal data analysis to identify and design a real-world healthcare solution. Integrated patient clinical data and retinal fundus images to develop a unified feature-level multimodal fusion network for early diabetes prediction.	Cognitive
CO2	Examine various problem domains through literature review, define the problems, and formulate the objectives for the capstone project.	PO4: Investigation	EP1 – Defining the Engineering Problem: Implemented state-of-the-art machine learning (MLP, Logistic Regression, Random Forest, CatBoost) and deep learning models (CNN – EfficientNet-B0) for analyzing clinical and image data.	Cognitive, Psychomotor

			Performed Exploratory Data Analysis (EDA), applied feature-level fusion, and used explainable AI techniques (Grad-CAM, SHAP) to ensure interpretability, robustness, and clinically meaningful predictions.	
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